Darwin and Evolution II

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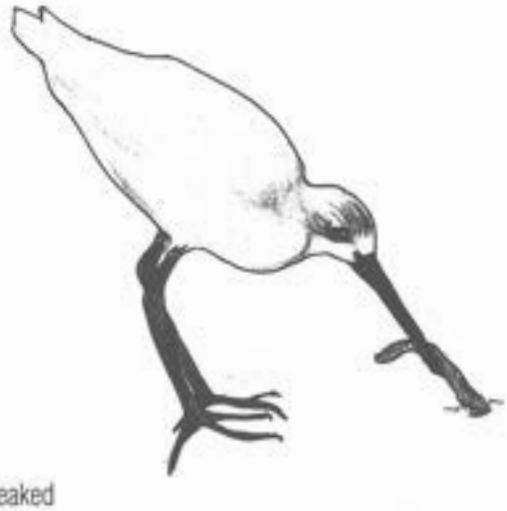
Various pressures can be applied on areas of land.

1 Decide which of the actions described below apply directional pressure on the piece of land described? Explain your answers.

- A certain naturally occurring plant species is removed from a paddock whenever it appears.
- Planting annuals (complete life cycle in one year) in a garden every summer.
- Weekly spraying of insecticide in an orchard.

Population: all the members of a particular species (related organisms) found in an area.

Consider a small population of 20 birds (10 male and 10 female). Due to a change in the environment their normal food sources (berries and caterpillars) are growing scarce. However, due to these changes, a new abundant food source is available (mud worms). In order to reach the worms the birds must have long, probing beaks. Long beaks are not common in this population, only four birds (two of each sex) have them.



2 Using this population as an example, show how a species may evolve. Use the following assumptions over three generations:

- five short-beaked birds die, or do not reproduce.
- the surviving birds reproduce
- successfully reproducing birds produce three young each
- long beaks are a dominant feature (only one dose of the gene is required for the offspring to express it)

3 What fact about the genomes (genetic material) of the four original long-beaked birds is unknown here?

4 With regards to your answer to Question 3, how might your evolution predictions be affected?

'Individuals can be selected but populations evolve.'

5 Discuss the statement above, with reference to the bird population mentioned previously.

Alfred Wallace

Working at about the same time as Darwin, Alfred Russel Wallace (1823–1913) came up with similar ideas about evolution. Wallace explained selective pressures using animal survival as his example. Those animals who can hunt more successfully, and cope better with difficulties in their environment, will survive illness and shortage of food, when others fail. By doing so they will live to reproduce, giving to their offspring the traits that made them better able to cope. This idea is now commonly known as 'survival of the fittest'.

6 Give a brief definition of 'trait'.

7 Give a brief definition of 'offspring'.

8 Briefly describe, by reference to a particular animal, an example of *survival of the fittest*.

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